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Submitted for the Practical Examination held on.....

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NAME: Rakshanaa. C STD: _____ SEC: C ROLL NO.: 188 SUB Software Constr-tio

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22.1.26

Ex: 1 Azure Devops Environment setup
(Create Epic, Features, User stories, task)

Aim:

To set up and access the azure devops environment by creating an organization through azure portal.

Installation:

1. Open your web browser and go to the azure website. Sign in using your microsoft account credentials. If you dont have a microsoft account you can create one.

2. Azure home page

3. Open devops environment in the azure platform by typing azure devops Organization in the search bar.

4. Click on my Azure Devops Organization link and create an Organization and you should be taken to the Azure Devops Organization Home page.

Result:

Successfully accessed the Azure Devops environment and create a new organization through the Azure portal.

29.1.26

Ex: 2 Azure Devops project setup and User Story Management

(create Epic, Features, User stories, tasks)

Aim:

To set up an Azure Devops project for Efficient collobration and agile work management.

1. Create an Azure account
2. Create the first project in your Organization

a. After the organization is set up, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's home page, click on the new Project button.

c. Enter the project name, description, and visibility options.

Name: choose a name for the ~~Pject~~ project (e.g, LMS)

Description: Optionally, add a description to provide more context about the project.

Visibility: choose whether you want the project to be private or public.

d. Once you've filled out the details, click create to set up your first project

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organization. you can switch b/w them from the top left corner. click on the organization name, and you should be taken to the Azure Devops Organization Home page.

4. Project Dashboard

5. To manage User Stories

a. From the left-hand navigation menu, click on boards. This will take you to the main Boards page, where you can manage work items, backlogs and sprints.

b. On the work items page, you'll see the option to add a work item at the top. Alternatively, you can find a + button or add New Work Item depending on the view you're in. From Add a work item dropdown, select User story. This will open a form to enter details for the new user story.

Result :

Successfully created an Azure Devops project with user story management and agile workflow setup.

5.2.26

EX: 3 Setting up Epics, Features and User Stories for project Planning.

Aim:

To learn about how to create Epics, User story, features, backlogs for your assigned project.

Create Epic, Features, User story, task:

1. Fill the Epic
2. Fill in Features of the Epic
3. Fill in User story of the Epic given.

Result:

Thus the creation of epics, features, user story and task has been created successfully.

12.2.26

EXP. 4 Sprint Planning

Aim:

To assign user story to specify sprints for the Music Playlist Batch creator project

Sprint planning:

Sprint 1
Sprint 2
Sprint 3
Sprint 4

Result:

The sprints are created for the music playlist Batch creator ~~proj~~ project.

19.2.26

Ex:5 Poker Estimation

Aim:

Create poker estimation for the user stories - Music Playlist Batch Creator Project.

Poker Estimation.

- In Azure Microsoft account click your main ongoing project
- click New and create a user story And add story points
- Each members in the team should add their story points in planning column

Ex:

member 1 : 3

member 2 : 5

member 3 : 8

(tester) QA : 8

Finally Every team members go with story point 5. According to poker Estimation.

Result:

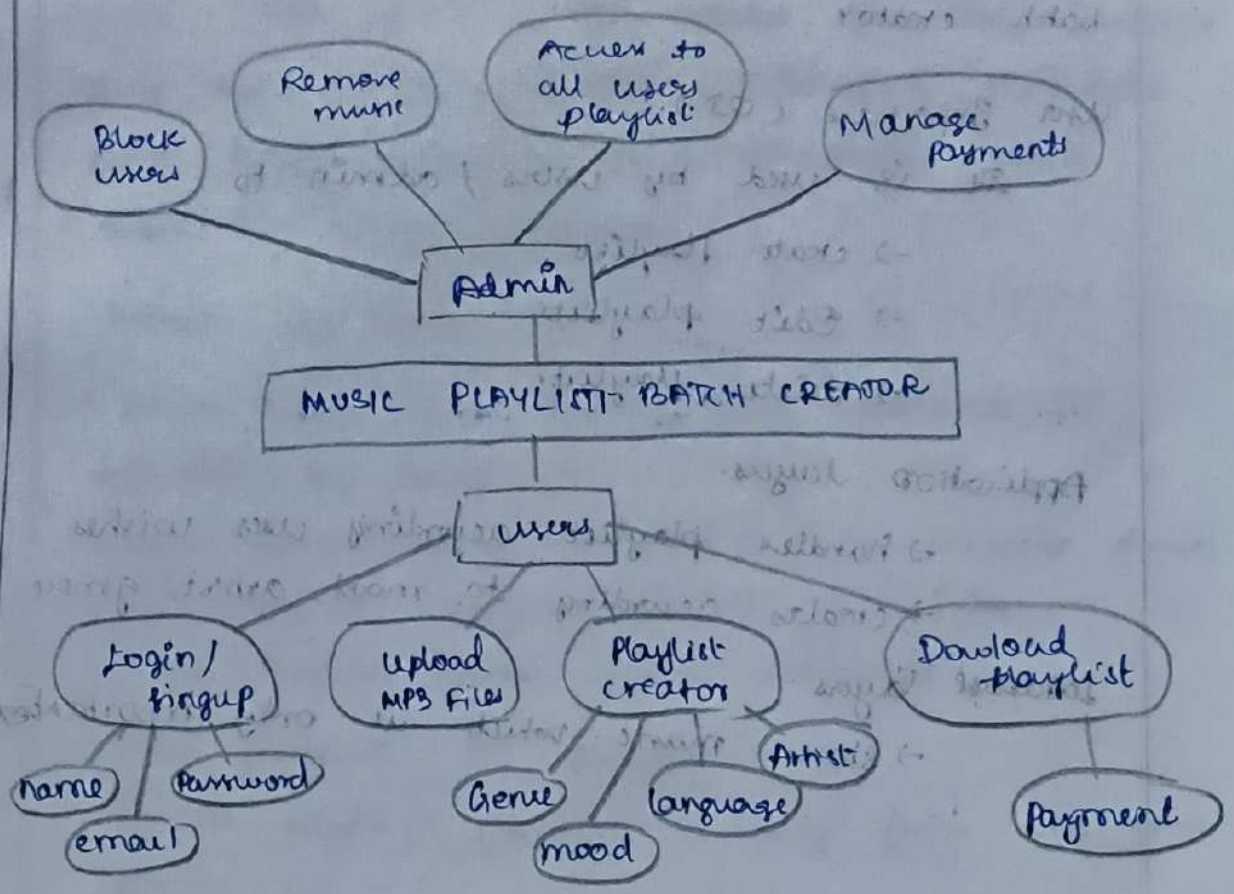
The Estimation of story points is created for the project using poker estimation.

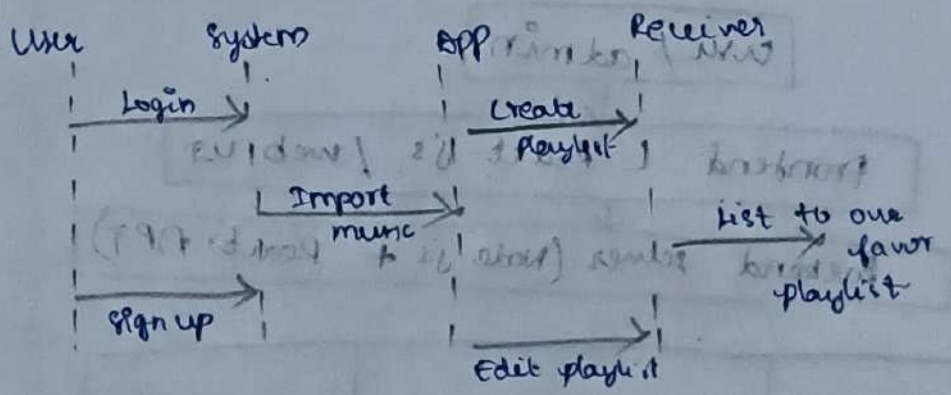
26.2.26

EX: 6 Designing class and sequence diagrams for project Architecture

Aim:

To design a class diagram & sequence diagram for the ~~app~~ Music playlist Batch creator.





Result:

Thus the class diagram and sequence diagram is drawn.

5.3.26

EX: 7 Designing Architecture and ER Diagrams for Project Structure.

Aim:

To design an architecture diagram & ER diagram for the music playlist batch creator.

User Interface (UI):

It is used by users / admin to

-> create Playlist

-> Edit playlists

-> delete playlists

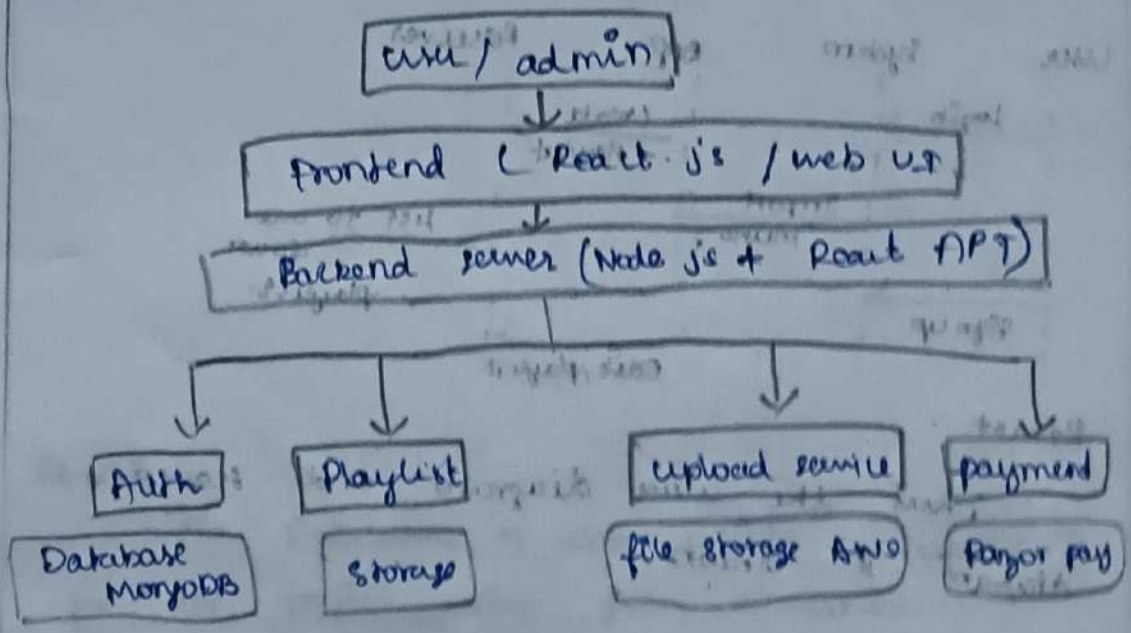
Application layers.

-> handles playlist according user wishes

-> creates according to mood, artist, genre

Database layers:

-> stores music which is only important



Result:

The architecture diagram is drawn successfully.

Experiment - 8

Testing (test plans & test case)

Aim:

To design test plans & test cases for the Music playlist batch Creator project.

Description:

Test plans were created for key features like login, playlist creation, editing & metadata. Both happy path & error scenarios were tested to ensure reliability.

Sample test cases:

- Login success: User enters valid credentials -> redirects to dashboard.
- Login Failure: wrong Password -> error message shown.
- Create Playlist: Playlist created successfully
- Empty Playlist Name: Error displayed.
- View playlist: All playlist show
- Offline Mode: Playlist loading fails with error

Result:

The test cases were successfully created & executed for major user stories ensuring proper functionality.

Experiment 9

Load testing & pipelines:

Aim:

To perform load testing & implement CI/CD pipeline for the project.

Description:

- Load testing was done using Jmeter to check performance under high traffic
- CI/CD pipeline was created using Azure Devops connected with GitHub.

Pipeline task:

- Code checkout
- Python environment
- Install dependenies
- Run test script

Outcome:

- Application handled multiple requests efficiently
- pipeline automated build & testing process

Result:

Load testing & pipeline setup were completed successfully for performance & automation.

Experiment - 10

GitHub Project structure & Naming :

Aim :

To organize the project structure & maintain proper naming conventions

Project structure :

- / Project → main App code
- / templates → HTML files
- / static → CSS, JS
- / tests → test cases
- requirements.txt → dependencies.
- app.py → main file

Naming conventions :

- Files : lowercase with underscores
  (playlist - manager.py)
- Variables : meaningful name
  (user - playlist)
- Functions : action - based
  (create - playlist ( ))

Result :

Project structure was organized clearly, improving reliability & maintability